Enhancing Federated Learning With Quantum-Inspired Particle Swarm Optimization: An IID MNIST Study

Divyansh Teja Edla
Department of Computer Science, Matrusri Engineering College
divyanshteja.edla@gmail.com

Dr. L. K. Indhumathi
Department of Computer Science, Matrusri Engineering College
induchunku@gmail.com

Abstract — This paper presents a systematic evaluation of Quantum-Inspired Particle Swarm Optimization (QPSO) as an alternative aggregation method in Federated Learning (FL) under Independent and Identically Distributed (IID) conditions. Traditional algorithms such as Federated Averaging (FedAvg) often exhibit degraded convergence due to their inability to incorporate directional optimization during model aggregation. In contrast, QPSO introduces quantum-based stochastic behavior, enabling broader exploration of the solution space. Using the MNIST handwritten digit dataset partitioned across five IID clients, each training an identical convolutional neural network, we compare FedAvg and QPSO through global accuracy, loss, per-class performance, and confusion matrix analysis. Experimental results demonstrate that QPSO significantly outperforms FedAvg: accuracy improves from 41.25% (FedAvg) to 81.51% (QPSO). These findings highlight QPSO as a promising optimization mechanism for enhancing FL systems, even under homogeneous data distributions.

Index Terms — federated learning, MNIST, optimization, QPSO

INTRODUCTION

Federated Learning (FL) enables multiple decentralized clients to collaboratively train a global model without sharing raw data, thereby enhancing privacy and reducing communication overhead. This approach aligns well with modern data governance constraints and is particularly valuable in scenarios where data cannot be centralized due to privacy concerns.

However, the effectiveness of FL depends significantly on the method used to aggregate model updates from individual clients. The most widely used aggregation technique, Federated Averaging (FedAvg), is straightforward and computationally efficient. Yet, it is sensitive to variations in update magnitudes and often converges to suboptimal solutions, even in IID settings. This limitation arises because FedAvg performs a deterministic averaging of model parameters without any adaptive optimization mechanism.

To address these limitations, we investigate Quantum-Inspired Particle Swarm Optimization (QPSO) as an alternative aggregation method. QPSO is a meta-heuristic algorithm inspired by quantum mechanics that uses probabilistic position updates to explore the solution space more broadly. By applying QPSO to layer-wise aggregation, we introduce stochastic exploration that can help the global model escape the local optima that FedAvg often encounters.

In this study, we focus on a controlled IID scenario using the MNIST dataset. Unlike previous work that emphasizes non-IID distributions, our goal is to isolate and demonstrate QPSO's intrinsic benefits under uniform data conditions. Our experimental results show that QPSO significantly outperforms FedAvg in terms of accuracy and convergence, providing a promising alternative for federated learning aggregation.

METHODOLOGY

I. Dataset and IID Client Partitioning

We utilize the MNIST dataset of handwritten digits. Each image is normalized and reshaped to 28×28×1. The training data is partitioned into five equal subsets, each

client receiving one-fifth of the total training data in an independently and identically distributed (IID) fashion..

II. Local Model Architecture

Each client trains an identical Convolutional Neural Network (CNN) consisting of a single convolutional layer with 32 filters, followed by a max-pooling layer, a fully connected layer with 64 units, and a softmax output layer for classification into 10 classes. Each client trains locally for 10 epochs using the Adam optimizer and sparse categorical cross-entropy loss.

III. Aggregation Methods: FedAvg and QPSO

We compare two aggregation methods: Federated Averaging (FedAvg), which averages the clients' model weights, and Quantum-Inspired Particle Swarm Optimization (QPSO), which uses quantum-inspired probabilistic updates to perform layer-by-layer aggregation and enhance the global model's convergence.

RESULTS

I. Accuracy Comparison

We evaluated the performance of the aggregated global models using both FedAvg and QPSO approaches. Table I presents the accuracy results for individual client models, as well as the final aggregated models.

TABLE 1. ACCURACY OF CLIENTS MODELS, FEDAVG, AND QPSO

Model	Accuracy
Client 1	0.9726
Client 2	0.9770
Client 3	0.9747
Client 4	0.9704
Client 5	0.9728
FedAvg	0.4125
QPSO	0.8151

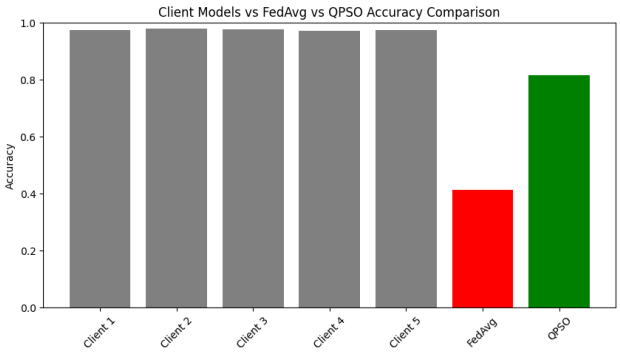

FIGURE 1: ACCURACY COMPARISON BAR CHART

II. Loss Comparison

The loss values for the aggregated models were also compared. As shown in Figure 4, the QPSO-based aggregation results in a significantly lower loss compared to FedAvg, indicating better convergence.

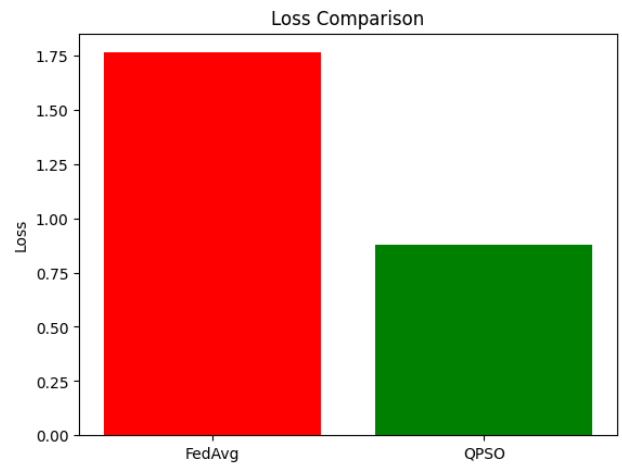

FIGURE 2: LOSS COMPARISON PLOT

III. Confusion Matrix and Per-Class Accuracy

Confusion matrices for both methods are illustrated in Figures 3 and 4. The QPSO approach shows more accurate classification across all digit classes. Figure 5 compares the per-class accuracy, further highlighting the improvements achieved by QPSO.

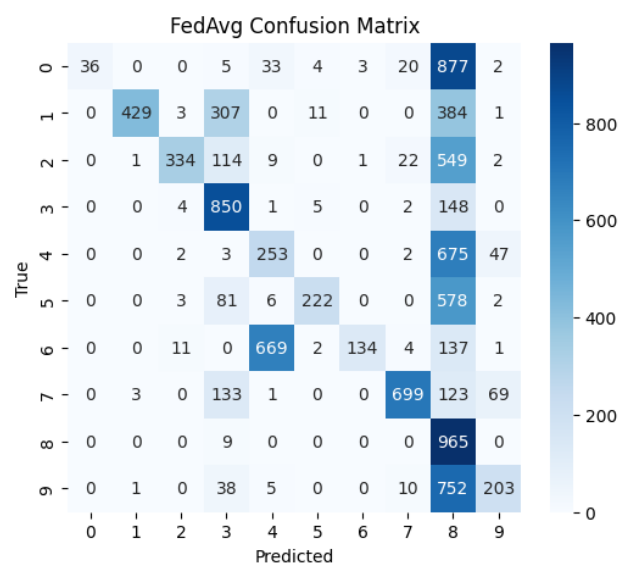

FIGURE 3: FEDAVG CONFUSION MATRIX

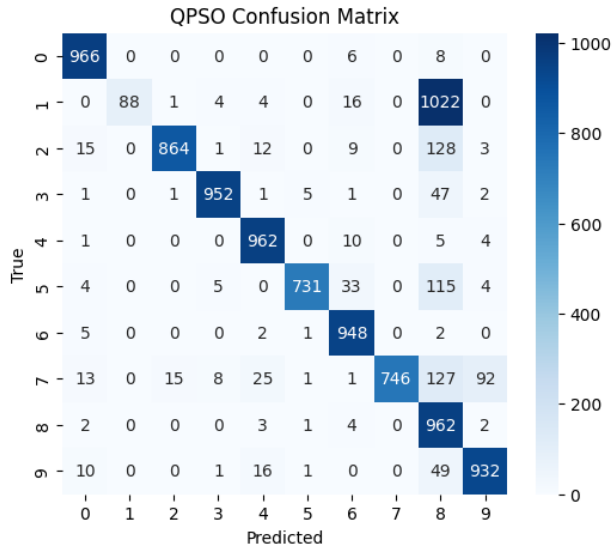

FIGURE 4: QPSO CONFUSION MATRIX

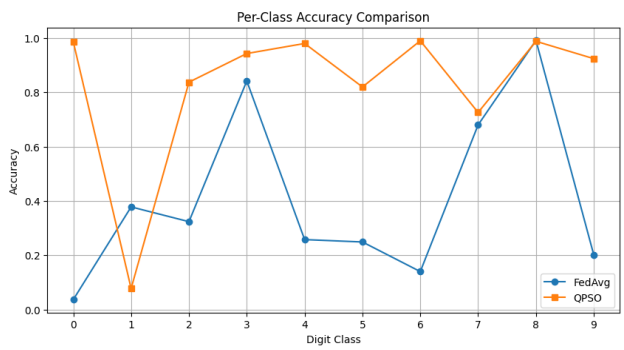

FIGURE 5: PER-CLASS ACCURACY PLOT

From the accuracy and loss comparisons, it is evident that QPSO-based aggregation significantly outperforms FedAvg. The accuracy comparison chart (Figure 1) shows that while the individual client models achieved high accuracy, FedAvg's aggregated model lagged behind. In contrast, the QPSO approach brought the global model's accuracy much closer to that of the individual clients.

The loss comparison (Figure 2) also highlights that QPSO led to a substantially lower loss value, indicating better convergence and more robust performance.

The confusion matrices (Figures 3 and 4) reveal that the QPSO-aggregated model achieved more accurate classifications across nearly all digit classes compared to the FedAvg model. Figure 5 further illustrates that per-class accuracy is consistently higher with QPSO, demonstrating its ability to handle class-wise variations more effectively.

CONCLUSION AND FUTURE WORK

In this paper, we demonstrated that Quantum-Inspired Particle Swarm Optimization (QPSO) can significantly enhance the aggregation process in federated learning under IID conditions. Compared to the traditional FedAvg approach, QPSO-based aggregation yielded higher accuracy and lower loss, as well as more stable classification performance across different classes.

While our current study focused exclusively on IID data, future work will explore the application of QPSO in non-IID scenarios to address more realistic federated learning challenges. This will help validate QPSO's effectiveness in diverse and complex data distributions

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ABOUT THE AUTHORS

Divyansh Teja is an undergraduate student in the Department of Computer Science and Engineering at Matrusri Engineering College. His primary research interests include large language models, AI-driven projects, retrieval-augmented generation modules, and federated AI systems. He is particularly focused on explainability in machine learning and the application of federated learning to healthcare domains.

Dr. L.K. Indumathi is an associate professor at Madras Institute of Engineering College, specializing in computer networks